



AERO4600 Automatic Flight Control Systems

Assignment 3 — 2024

Ben Miller^c Jabez Mehanathan^c Kaitlyn Ashcroft^c

^a *Student of Aerospace Systems Engineering,
The University of Newcastle, Callaghan, NSW 2308, AUSTRALIA
Student Number: c3328484
E-mail: c3328484@uon.edu.au*

^b *Student of Aerospace Systems Engineering,
The University of Newcastle, Callaghan, NSW 2308, AUSTRALIA
Student Number: c3355997
E-mail: c3355997@uon.edu.au*

^c *Student of Aerospace Systems Engineering,
The University of Newcastle, Callaghan, NSW 2308, AUSTRALIA
Student Number: c3350754
E-mail: c3350754@uon.edu.au*

Abstract

This Report examines the lateral dynamics and gust response of an aircraft and builds on prior work ?, focusing on utilizing Matlab System Identification Toolbox as well as simulating aircraft response to gust inputs. The primary lateral modes analysed in this report are the side-slip (β) and bank angle (ϕ) of the aircraft, and are evaluated through the influence of aileron/rudder control inputs and gust disturbances. This report also evaluates the impact of sensor noise on the the system identification model and methods of improving the robustness of the system identification process. Additionally, the gust/turbulence was modelled using the Dryden Spectra method to aid in simulating the gust responses, in order to compute the maximum side-slip and bank angle perturbations. These finding contribute to modelling and understanding the aircraft's dynamics and control in the lateral motion.

Contents

1. Introduction	4
2. Q1 – Vertical Speed Transfer Function	4
3. Q2 – Pitch Rate Autopilot	4
4. Q3 – Vertical Speed Guidance Loop	5
4.1. Loop Topology	5
4.2. Aircraft Response to Vertical Speed Command	5
4.3. Pitch Rate Response	6
4.4. Elevator Activity	7
5. Q4 – Design Auto-Throttle Control System	8
5.1. MIMO Loop Topology	8
5.2. Guidance Loop Topology Specifications	8
5.3. Primary Control Effects	8
5.4. Cross-coupling Control Effects	10
5.5. Elevator and Throttle Response	10
5.6. Auto-Throttle Effectiveness	10
6. Final Design	10
6.1. Q5 – Implementation of the control and guidance laws	10
6.2. Q6 – Gust Effectiveness	10
6.3. Q7 – Commanded Large Vertical Speed	10
7. Conclusion	10
A. Example of a Table	11

1. Introduction

test here

2. Q1 – Vertical Speed Transfer Function

To assess the feasibility of using the aircraft elevator (δ_e) to directly control the vertical speed (v_s), the transfer function $G_{\delta_e}^{v_s}$ was determined. Where $C = [0 \ 0 \ 0 \ 1 \ 0]$, the Transfer Function for v_s was found to be (2.1).

$$G_{\delta_e}^{v_s}(s) = \frac{8.57(s - 16.89)(+12.79)(s + 0.04162)}{(s^2 + 0.03407s + 0.06989)(s^2 + 4.67s + 10.92)} \quad (2.1)$$

The stability of aircraft's control system can be determined by the location of its poles, shown in figure 1. Poles can be seen on the right half of the s-plane, demonstrating instability.

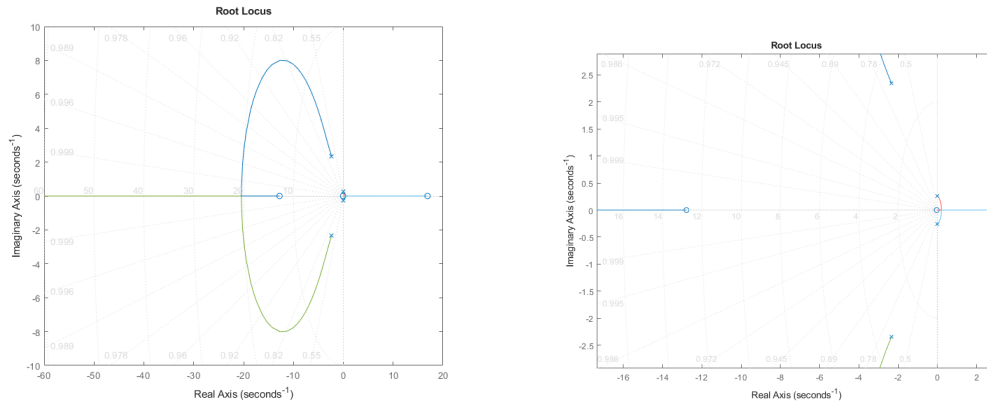


Figure 1: Root Locus of $G_{\delta_e}^{v_s}(s)$

3. Q2 – Pitch Rate Autopilot

To design a pitch rate autopilot which controls

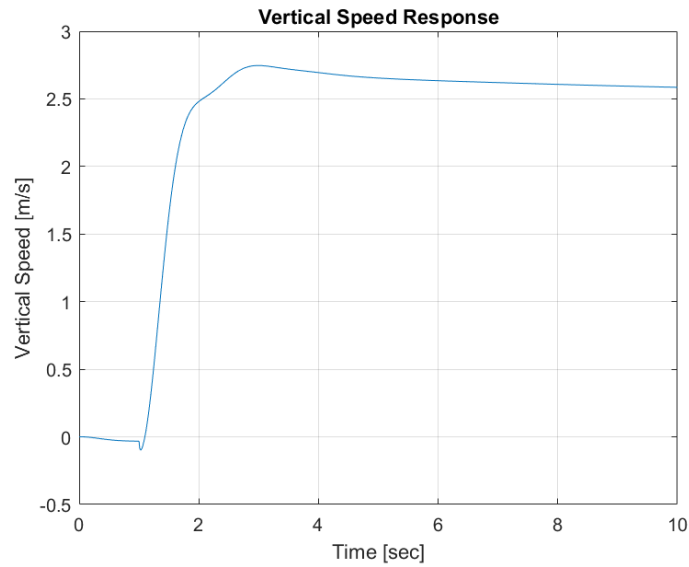


Figure 3: Vertical Speed Response

the reference value, achieving stability within approximately 10 seconds. This response indicates that the system meets the rise time and settling time requirements, effectively balancing speed of response with controlled damping. The smooth convergence to the command confirms the system's capability to achieve accurate vertical speed tracking.

4.3. Pitch Rate Response

The pitch rate response, shown in Figure 4, reflects the performance of the inner pitch rate controller.

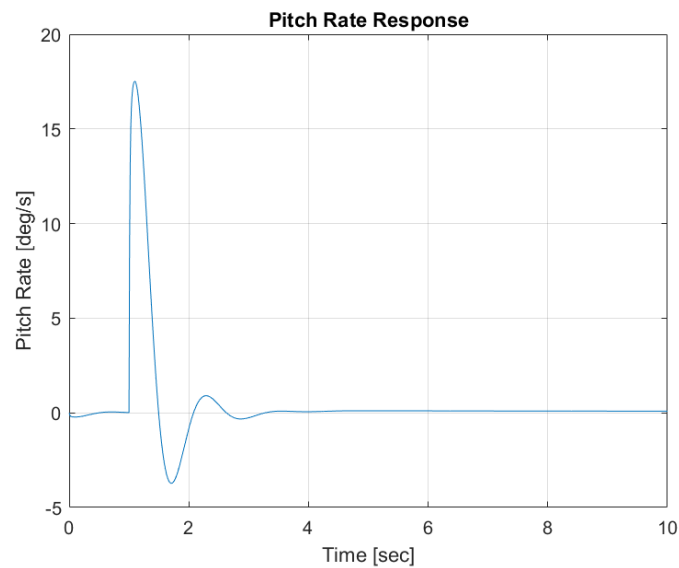


Figure 4: Pitch Rate Response

The response shows a peak reaching 16deg/s , followed by a quick correction with minor oscillations before settling to zero. This peak and correction indicate the controller's rapid adjustment to the pitch rate required to meet the vertical speed command of 500ft/min (2.54m/s), as outlined in Section 4.2. These damped oscillations demonstrate the pitch rate controller's stability and responsiveness.

4.4. Elevator Activity

Figure 5 displays the control activity of the elevator in response to the inputted vertical speed command.

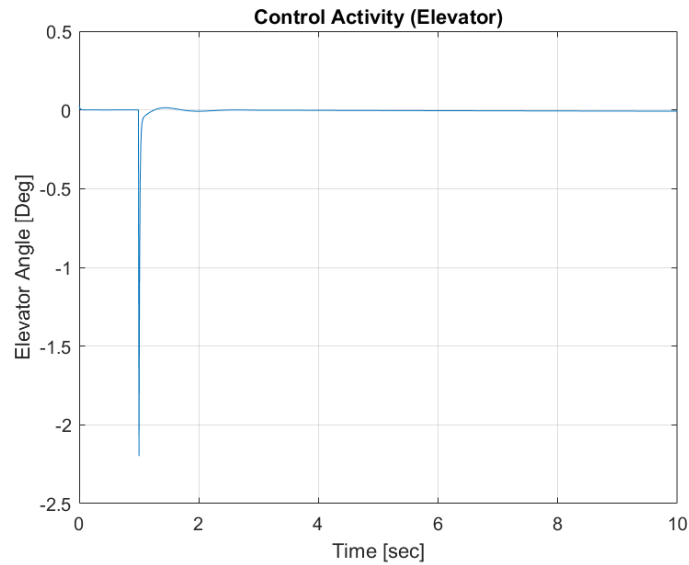


Figure 5: Elevator Control Activity

The elevator initially deflects sharply to approximately -2.3degrees as the controller adjusts to achieve vertical speed command, followed by the elevator deflection stabilizing back to zero, demonstrating its minimal effort required once the target vertical speed is reached.

5. Q4 – Design Auto-Throttle Control System

In this section an auto-throttle design was attempted, wherein errors in airspeed would determine throttle command.

5.1. MIMO Loop Topology

The auto-throttle system was to be achieved with the MIMO loop topology shown in Figure 6.

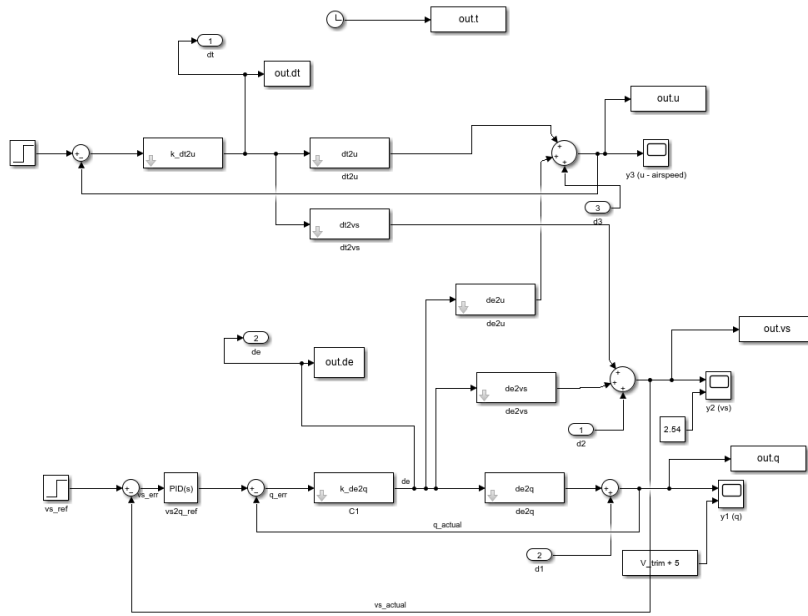


Figure 6: MIMO Simulink Loop for Auto-Throttle Design

5.2. Guidance Loop Topology Specifications

This loop was designed to the same specifications as the vertical speed guidance loop, with a vertical speed step input of 500 ft/min. The other constraints could not be implemented however due to difficulties in controlling the system. In addition to the vertical speed input there was also an input for u (airspeed).

5.3. Primary Control Effects

The primary control effects, namely the transfer function of throttle to airspeed ($(dt2u)$), and elevator to vertical speed ($de2vs$), are shown in 7 and 8.

These effects tend to follow a more sinusoidal pattern, reaching similar magnitudes over the simulation period. This is quite different from the responses found from solely the vertical speed control loop (in 4.1). This indicates the the controls have more of an effect on the outputs over time, although

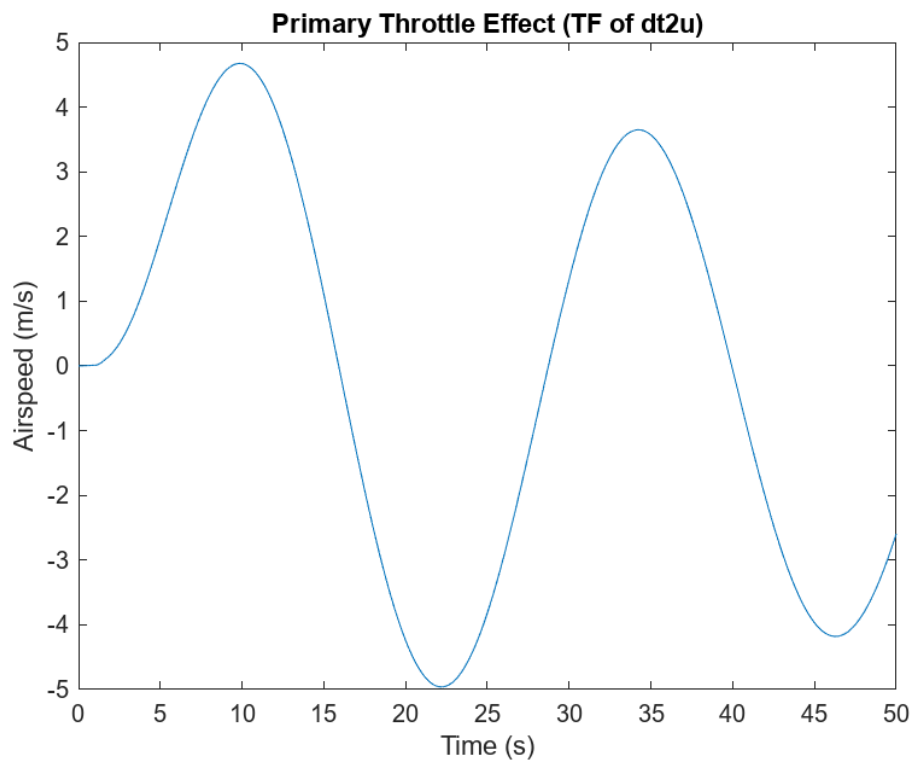


Figure 7: Primary Control Effect: Throttle to Airspeed

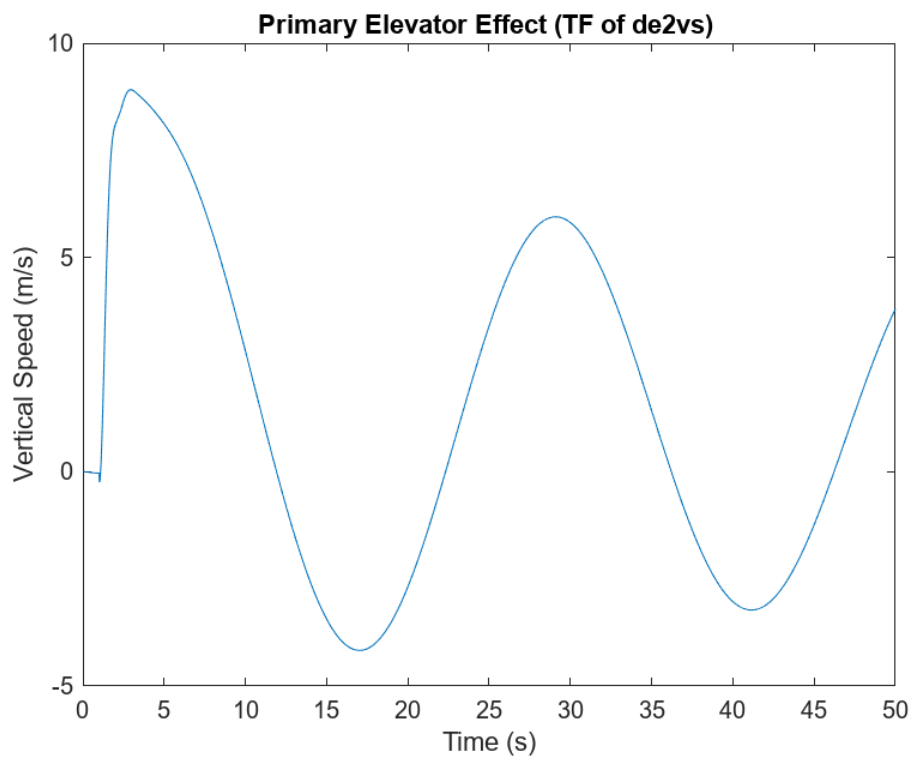


Figure 8: Primary Control Effect: Elevator to Vertical Speed

because of the increased complexity of the system it is more difficult to tune, and does not reach a stable point (like the outputs of the vertical speed control loop).

5.4. Cross-coupling Control Effects

The cross-coupling control effects, namely the transfer function of throttle to vertical speed ($\frac{dv}{du}$), and elevator to airspeed ($\frac{du}{\delta e}$), are shown in ?? and ??.

5.5. Elevator and Throttle Response

5.6. Auto-Throttle Effectiveness

6. Final Design

6.1. Q5 – Implementation of the control and guidance laws

6.2. Q6 – Gust Effectiveness

6.3. Q7 – Commanded Large Vertical Speed

7. Conclusion

References

A. Example of a Table